

React Restaurant Web Application Report

Abstract

This project is a React-based restaurant website that provides a modern user interface for browsing pages such as Home, About, Menu, and Contact. It uses React Router for navigation and component-based architecture to improve reusability and scalability. The system simulates a fine dining restaurant experience including booking functionality and contact forms.

System Design

The application follows a single-page application (SPA) architecture using React. It is structured into reusable components such as Navbar and multiple pages (Home, About, Menu, Contact). React Router handles navigation between pages without reloading the page. The Home page includes a booking modal implemented using a checkbox toggle method. Forms are used in Contact and Booking sections for user input.

Technologies Used

- React.js for UI development - React Router DOM for routing - JavaScript (ES6+) - HTML5 & CSS3
- CSS Variables and Media Queries for styling and responsiveness - Vite (assumed build tool)

Code Snippets

App Routing Example:

```
import { Routes, Route } from "react-router-dom";

<Routes>
  <Route path="/" element={<Home />} />
  <Route path="/about" element={<About />} />
  <Route path="/menu" element={<Menu />} />
  <Route path="/contact" element={<Contact />} />
</Routes>
```

Navbar Component:

```
import { Link } from "react-router-dom";

<nav>
  <ul>
    <li><Link to="/">HOME</Link></li>
    <li><Link to="/about">ABOUT</Link></li>
    <li><Link to="/menu">MENU</Link></li>
    <li><Link to="/contact">CONTACT</Link></li>
  </ul>
</nav>
```

Home Booking Modal:

```
<input type="checkbox" id="book-toggle" />
<label htmlFor="book-toggle">book a table</label>
<div className="modal">
  <form>
```

```
<input type="text" placeholder="Full Name" />
<button>Confirm</button>
</form>
</div>
```